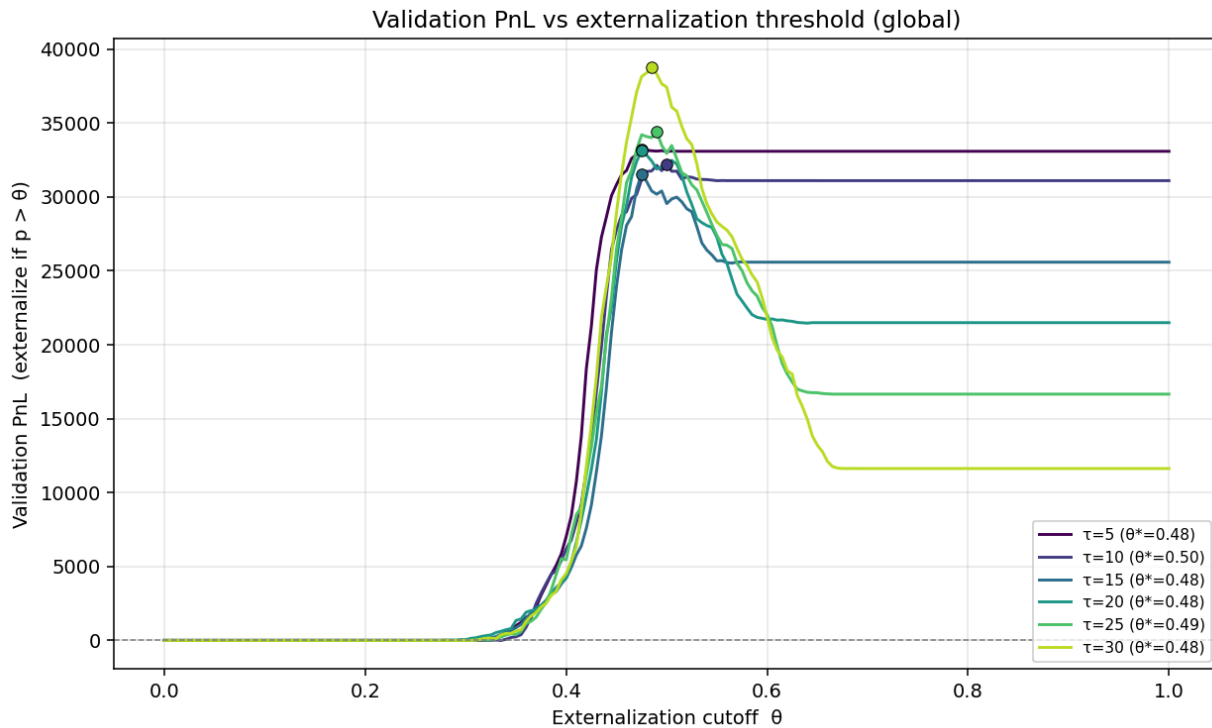


Task 4: Optimal Externalization Threshold

Using the Task-3 model, a trade is externalized (netted to zero PnL) iff its predicted adversity p exceeds a cutoff θ . Internalized trades realise the value side $\cdot V \cdot (Mt-TP)$. We sweep $\theta \in [0,1]$, pick θ^* maximising validation PnL, and report held-out test PnL.



The validation curves are single-peaked and the optimum sits near $\theta^* \sim 0.47-0.50$ for every horizon, that is, keep trades the model rates below $\sim 50\%$ adverse, externalize the rest.

Discussion:

- What happens when θ is too low (over externalization)? What revenue do you lose?
Ans: we externalize even low-probability trades, throwing out even the positive spread/PnL of benign flow. As $\theta \rightarrow 0$ essentially everything is pushed out (every trade is externalised) and PnL collapses toward zero. We lose the revenue that makes market-making worthwhile (zero risk zero reward).
- What happens when θ is too high (under-externalization)? What adverse flow do you absorb?
Ans: we keep high-probability-adverse trades and take their losses. As $\theta \rightarrow 1$ PnL falls back to the unhedged 'keep-everything' level, which is sharply negative for toxic flow (clients E, F at long horizons). θ^* balances forgone spread against absorbed adverse flow.

- Is the optimal θ^* the same across all clients, or should it be client-specific? Justify with evidence.

Ans: Client-specific thresholds (one θ^* per client per horizon) is better than the global rule on validation and is comparable on test (table below). The reason is structural. Benign clients (A-C) are best served by a high threshold that keeps almost all flow, whereas for toxic clients the per-client optimum collapses to $\theta^*=0$ at long horizons, that is, externalize everything, because any retained flow loses money. A single global θ cannot express both. I therefore recommend client-specific thresholds, while noting the usual that very thin/toxic clients risk in-sample overfit, so the test numbers are the honest estimate of realised performance. (in the table CS means client specific)

tau	global theta*	val PnL	test PnL	val PnL (CS)	test PnL (CS)
5	0.475	33179	39643	33234	39669
10	0.5	32181	38127	33256	37108
15	0.475	31558	35395	32911	37611
20	0.475	33114	37867	34882	38795
25	0.49	34412	42665	36898	42655
30	0.485	38755	45428	39897	45463

Final held-out test PnL with the chosen thresholds ranges from ~35k ($\tau=15$) to ~45k ($\tau=30$); per client and τ thresholds and test PnL are in task4_results.csv.